



Clinical Physiology

PTH6111.01 (Summer 2025)
Dept of Physical Therapy

Instructor Info —



Sean Collins, PT, ScD



Office Hrs: MW 1-3pm



SRH (Hall) 211



Canvas



smcollins1@plymouth.edu

Course Info —



Credit Hours: 3



Mon, Weds, Friday



11a-12:15p



HPC W210

About —

Semester Clock Hours: 40

Weekly Clock Hours: 4.5

Prerequisites: DPT Matriculation

Corequisites: PTH6110, PTH6112,
PTH6115, PTH6116

Mode of Delivery: In Person

Learning Activities: Lecture,
Discussion

Lab Component: 3 hours of class time

Course Description

Clinical Physiology covers the function of the muscular; and structure / function of the neuromuscular, circulatory, pulmonary, endocrine, renal and metabolic systems. Students relate anatomical and physiological systems to functional capabilities. Students apply physiology to reasoning about clinical scenarios and apply concepts to patient/client problems with the WHO International Classification of Function for system organization.

Student Learning Outcomes

1. Describe and discuss the anatomy and function of the cells and cellular components of the muscular, circulatory, pulmonary, endocrine, renal, and metabolic systems. (7A)
2. Describe the development of the musculoskeletal, circulatory, pulmonary, endocrine, renal and metabolic systems. (7A)
3. Describe and analyze the different energy systems (ATP/CP, glycolysis, oxygen transport/ electron transfer) used in different activities (eg, one minute, anaerobic, or aerobic activities). (7A)
4. Analyze systematic interrelationships between the muscular, circulatory, pulmonary, endocrine, renal and metabolic systems in functional activities. (7A)
5. Identify possible causes of perturbations to homeostasis at rest and with physiological challenges (physical stress, nutritional stress, environmental stress). (7A, 7C)
6. Relate system function and capacity with multi system function and capacity. (7A, 7C)
7. Demonstrate the ability to apply the basic concepts of physiology to the analysis of patient/client problems related to the muscular, circulatory, pulmonary, endocrine, renal and metabolic systems. (7A, 7C)
8. Relate the the International Classification of Function (ICF) to describe how physiological systems emerge as function and consider the whole system impact of dysfunction (impairments) (7D21)
9. Show how to perform cardiovascular/pulmonary tests and measures including: Heart rate, Respiratory rate, pattern and quality, Blood pressure, Pulse Oximetry in an isolated practice condition. (7D19)

Course Materials

Required Texts

Collins SM. Clinical Physiology: A muscle centered approach. Manuscript in Preparation, 2024.

Other

Link to: Clinical Physiology Course Companion GPT
Additional readings available on Canvas.

References

All references utilized in the book and in the class are available in an online Zotero group library (if you would like access you need to join Zotero (free) and request to be a member of the group) at this link: Clinical Physiology Zotero Group

Grading

There are a total of 100 points available in this class. These points will come from the following areas.

60 points	Quizzes (5)
35 points	Cumulative Final Exam
5 points	Vital Sign Competency Check
Adjustment	Professional Conduct (See below)

Grading Scale

A $\geq$ 94
90 $\leq$ A- < 94
87 $\leq$ B+ < 90
83 $\leq$ B < 87
80 $\leq$ B- < 83
77 $\leq$ C+ < 80
73 $\leq$ C < 76
F < 73

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 10 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C).

Assignment Descriptions

Quizzes (60 points total)

- Quiz 1 (10 points) Chapters 1 & 2 (Introduction & Fundamentals)
- Quiz 2 (20 points) Chapters 3 - 6 (Muscle Function)
- Quiz 3 (10 points) Chapters 7 & 8 (Microcirculation & Renal Clearance)
- Quiz 4 (10 points) Chapters 9 & 10 (Circulation & Cardiac Pump)
- Quiz 5 (10 points) Chapters 11 & 12 (Respiration & Ventilation)

Quizzes occur throughout the semester (see the schedule for the exact dates). The allotted time is 72 seconds per question and the number of questions will be announced prior to the quiz. The remaining class time will be used to review and discuss the quiz.

Cumulative Final Exam (35 points total)

- Part 1 based on material from Quiz 1: 5 points towards final exam
- Part 2 based on material from Quiz 2: 5 points towards final exam
- Part 3 based on material from Quiz 3: 5 points towards final exam
- Part 4 based on material from Quiz 4: 10 points towards final exam
- Part 5 based on material from Quiz 5: 10 points towards final exam

Quiz Grade Replacement

Quiz grade replacement, also known as the final exam priority, is that grades on Quiz 1 - 5 can be replaced by a higher score achieved on the respective part of the Final Exam (Parts 1 - 5, respectively). For example, if you earned 6/10 points in Quiz 1, but earned 4/5 points in Part 1 of the Final Exam, your Quiz 1 grade will change to 8/10. This course policy encourages students to continue to improve their understanding throughout the semester and to reflect on that final state of understanding in the final grade. It also negates the need for makeup quizzes throughout the term.

Vital Sign Competency Check (5 points total)

Students will demonstrate competency in the practical skills learned in the two lab sessions of this course (cardiovascular and pulmonary vital signs). There are two lab sessions in the final two class meetings. The first session demonstrates and provides an opportunity to practice the skills. The second session provides an opportunity for each student to demonstrate competency in the skill. If competency is not met in this session, the student must arrange for a competency check before the end of the term.

Makeup Quizzes

There will be no makeup quizzes (see above, including the Quiz Grade Replacement section). A makeup final exam will be given only in specific cases of absence (documented absences). If possible, circumstances that may warrant absence should be discussed with the instructor prior to the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Professional Conduct

Students are reminded of the program expectation that they follow Professional Conduct, see the Student Handbook, Professional Conduct and Reasonable Professional Expectations for the process of handling violations, remediation and consequences at the program level. Violations of Professional Conduct (such as unexcused absences) will result in a verbal warning. Continued violation will result in a written notification of a grade reduction of up to 10%. Violations of Professional Conduct that reach an egregious level, such as cheating on a quiz or exam, will result in immediate failure of that quiz or exam.

Program & University Policies & Information

Doctor of Physical Therapy Student Handbook

Students are reminded that more detailed information on program policies, including the appropriate and ethical use of Artificial Intelligence (AI), can be found in the Plymouth State University Doctor of Physical Therapy Student Handbook.

Academic Standing

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 9 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C). Given the Make-up examinations will be given only in specific cases of absence (documented absences) for the Final Exam. Circumstances that may warrant an absence should be discussed with the instructor before the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Fair Grading Policy

Fair and equitable grading reflects values to which all members of the Plymouth State community commit themselves. Grades are used to assess the relative extent to which students achieve course objectives in all for-credit courses at PSU. Academic freedom allows instructors to determine course objectives within the bounds of established curricula, and the means by which a student's mastery of those objectives will be evaluated; and evaluate the quality of work on individual exams or assignments. Information can be found at <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Fair Grading.

Graduate Academic Policies

The DPT Program is a selective admissions cohort-structured program. Some Graduate Academic Policies may not be relevant to the program. Students should refer to the Doctor of Physical Therapy Program Handbook for specific information about Scheduling, Capstones, and Internships. The sections on Grading System and Academic Standing Policy do apply to DPT students: <https://coursecatalog.plymouth.edu/university-policies-procedures/graduate-academic-policies/>

Student Support

Student Support Foundation (SSF) provides short-term emergency financial assistance and long-term student support. For more information, see: <https://campus.plymouth.edu/student-support-foundation/> Student Support Foundation also runs a food pantry, located in Belknap Hall. To learn more about SSF or access the food pantry, either via open hours or a private appointment, contact the SSF advisor, at psu-ssf@plymouth.edu

Academic Dishonesty

Cheating and plagiarism are very serious forms of academic dishonesty. Any use of unauthorized assistance on exams, papers, homework assignments, or other course work constitutes cheating. Knowingly providing assistance during an exam or allowing other students to copy one's work is also a serious form of academic dishonesty. Plagiarism consists of submitting written work that has been developed wholly or partially by someone else. Submitting written work in which the ideas of others have been duplicated or even paraphrased without proper reference to the author is also a form of plagiarism. Also considered plagiarism is the acquisition of term papers or other assignments from another source and subsequent presentation of these materials as the student's own work. In addition, students may not use papers in more than one course without the permission of both instructors.

Any student suspected of academic dishonesty will be reported to the Program Director immediately. A full description of academic integrity, violations and procedures when a violation to academic integrity in a course is suspected can be found at this website under Academic Integrity.

Attendance Policy

Attendance is expected at all class sessions. Students are ultimately responsible for their attendance, however faculty will assess students Professional Expectations based, in part, on their attendance (See Professional Conduct under Student Evaluation/Grading). Students who miss class for any reason are responsible for the information and work missed. Students should speak with the professor before missing the scheduled class when able.

Campus Accessibility Policy

Plymouth State University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability that requires accommodations, you should immediately contact Campus Accessibility Services (CAS) in Speare 210 (535-3300) to determine whether you are eligible for such accommodations. Academic accommodations will only be considered for students who have registered with CAS. If you have a Letter of Accommodation for this course from CAS, please provide the instructor with that information privately so that you and the instructor can review those accommodations: <https://campus.plymouth.edu/accessibility-services/>

LibGuide for Physical Therapy:

The Libguide for Physical Therapy is an information sharing platform created through a collaboration between the library and the DPT faculty that presents relevant resources to support students in the Physical Therapy program. Examples of information you will find include tutorials to better understand citing sources in APA and AMA style, resources on how to avoid plagiarism, quick access to relevant databases and interlibrary loan as well as a contact link to the programs' assigned librarian. To view the DPT Program Libguide go to <https://library.plymouth.edu/physical-therapy>.

Cell Phones

Cell phone use (including texting) is prohibited in class. Also, please respect your classmates by turning your phone to vibrate prior to class start time. Do not turn phones off as the University uses texts to relate emergency information.

Diversity and Inclusivity Statement

I consider this classroom to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. See this web page for the Plymouth State University IDEA Center Inclusion, Diversity, Equity, and Access.

Tentative Class Schedule

Note: As changes arise they will be discussed in class.

There is a quiz debrief following each quiz.

Week of	Monday	Weds	Friday
June 2	Chp 1 - Introduction	Chp 2 - Fundamentals	Chp 2 - Fundamentals (cont)
June 9	Quiz 1: Chps 1 & 2	Chp 3 - Tension	Chp 4 - Excitation
June 16	Chp 5 - Regulation	Chp 6 - Energetics	Muscle Function Overview
June 23	Quiz 2: Chps 3-6	Chp 7 - Microcirculation	Anatomy Lecture (Dr. Duffy)
June 30	Chp 8 – Renal Clearance	Chp 8 – Renal (cont)	No Class – July 4 th
July 7	Quiz 3 – Chps 7 & 8	Chp 9 – Circulation	Chp 9 (cont) & Chp 10 Cardiac Pump
July 14	Chp 10 – Cardiac Pump	Quiz 4 – Chps 9 & 10	Chp 11 - Respiration
July 21	Chp 11 - Respiration (cont)	Chp 12 - Ventilation	Chp 12 - Ventilation (cont)
July 28	Quiz 5 – Chps 11 & 12	Vital Signs Lab	Vital Signs Lab
August 4	Final Exam & OSCE as Scheduled		

Mea Culpa, Veniam Peto

Mea Culpa, Veniam Peto is Latin for my fault, and I ask for forgiveness. It is an acknowledgment of a personal error or fault and a request to accept my apologies for any errors in this syllabus. They are bound to occur. Please bring any errors you find to my attention as soon as possible.